Solution

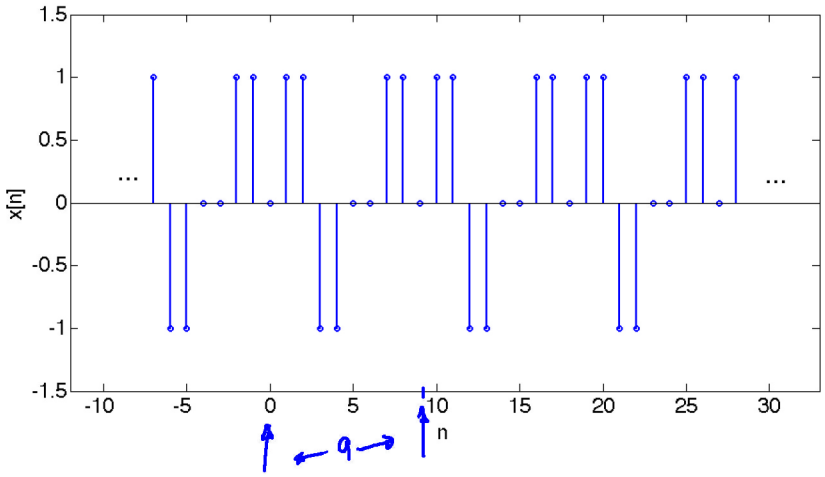
Signals: The Basics

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1. Consider the following signal.



This is a

a) discrete-time signal

b) continuous-time signal

and is

a) non periodic

b) periodic with fundamental period $N = 4$

c) periodic with fundamental period $N = 5$

d) periodic with fundamental period $N = 6$

e) periodic with fundamental period $N = 7$

f) periodic with fundamental period $N = 8$

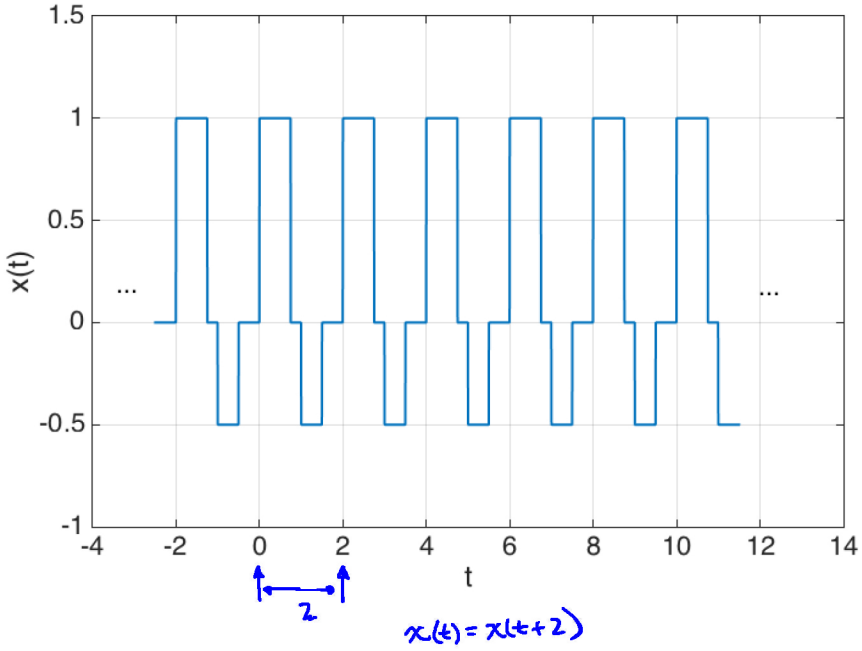
g) periodic with fundamental period $N = 9$

h) periodic with fundamental period $N = 10$

i) periodic with fundamental period $N = 11$

j) periodic with fundamental period $N = 12$

2. Consider the following signal.



This is a

a) discrete-time signal

b) continuous-time signal

and is

a) non periodic

b) periodic with fundamental period 1 second

c) periodic with fundamental period 2 second

d) periodic with fundamental period 3 second

e) periodic with fundamental period 4 second

f) periodic with fundamental period 5 second

g) periodic with fundamental period 6 second

h) periodic with fundamental period 7 second

i) periodic with fundamental period 8 second

3. The signal $x[n]$ is defined as

$$x[n] = \sin^2(\pi n/5)$$

This signal is

a) non periodic

b) periodic

periodic if there is some integer N so that $x[n] = x[n+N]$ for all n

$$x[n] = \sin^2(\pi n/5) = \sin^2(\pi n/5 + \pi) = \sin^2(\pi(n+5)/5) = x[n+5]$$

4. The signal $y[n]$ is defined as

$$y[n] = \cos^2(\pi n/6)$$

Find the fundamental period of $y[n]$.

Smallest N so that $y[n] = y[n+N]$

$$y[n] = \cos^2(\pi n/6) = 0.5 + 0.5\cos(2\pi n/6)$$

smallest phase shift that does not change cos

$$= 0.5 + 0.5\cos(2\pi n/6 + 2\pi)$$

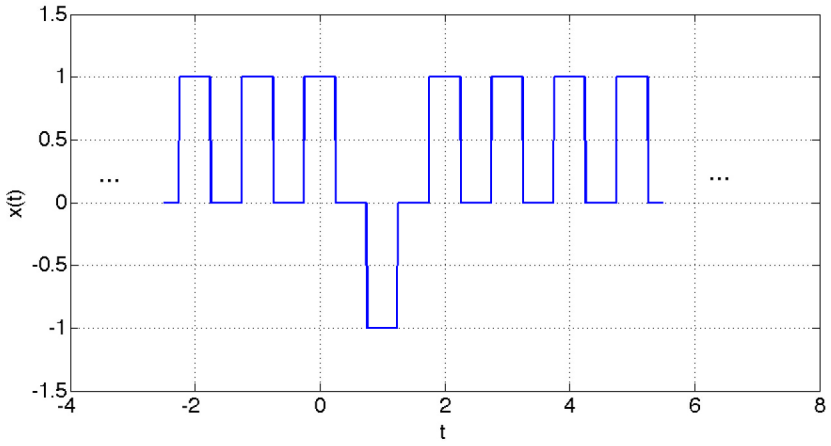
$$= 0.5 + 0.5\cos(2\pi(n+6)/6)$$

$$= \cos^2(\pi(n+6)/6)$$

$$= y[n+6]$$

so N = 6 this is the smallest or fundamental period as we used the smallest phase change

5. Consider the signal $x(t)$ shown below. This signal is



doesn't repeat for all t
0 < t < 2 is unique

a) non periodic

b) periodic with fundamental period 1 second

c) periodic with fundamental period 2 second

d) periodic with fundamental period 3 second

e) periodic with fundamental period 4 second

f) periodic with fundamental period 5 second

g) periodic with fundamental period 6 second

h) periodic with fundamental period 7 second

i) periodic with fundamental period 8 second

6. Use MATLAB code CTSinusoids.m to generate a graph of a sinusoid

$$x(t) = A\cos(2\pi f t + \phi)$$

on $0 \le t \le 10$ for the following values of A, f, and ϕ. In each case find the fundamental period.

a) $A = 2,\ f = 0.5,\ \phi = 0$

b) $A = 4,\ f = 1.25,\ \phi = 0$

c) $A = 4,\ f = 0.5,\ \phi = \pi/4$

d) $A = 2,\ f = 0.5,\ \phi = \pi/2$

$x(t) = x(t+T)$ for some T and all t

$x(t) = A\cos(2\pi f t + \phi)$

$= A\cos(2\pi f t + \phi + 2\pi)$

$= A\cos(2\pi f(t + \frac{1}{f}) + \phi)$

$= x(t + \frac{1}{f})$

so $T = \frac{1}{f}$

independent of A, ϕ

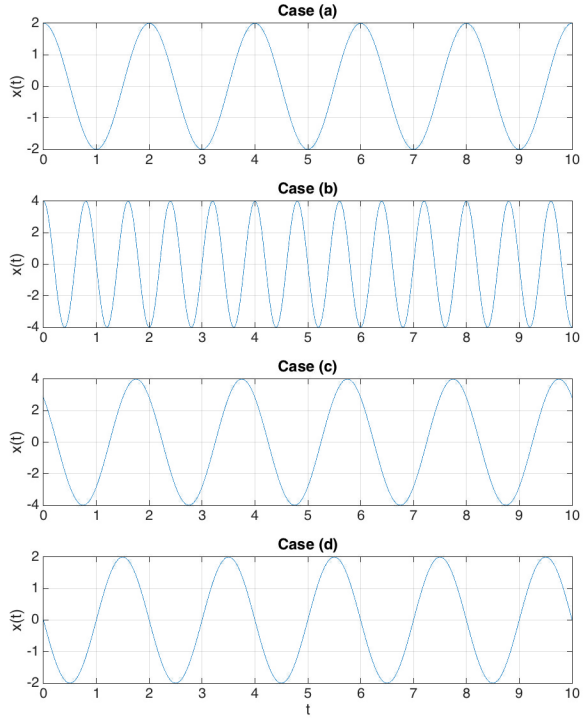
a) $T = 2$ sec

b) $T = 0.8$ sec

c) $T = 2$ sec

d) $T = 2$ sec

graphs on following page


Case (a)
Case (b)
Case (c)
Case (d)
x(t)
t